Two ceilings moved at once

N=96 and N=48, 2026-10-26

N=96: the ceiling was TRAINING DATA, not the learning rate.

4x data takes the shared pool from 0.8267 to 0.9273 [0.9131, 0.9394] -- the Wilson floor clears 0.90. Cosine lr helps too (+0.0473) but only half as much, and it does not clear.

I had that diagnosis backwards, and said so in advance.

Finding 15 argued the plateau was optimization because train sat 0.0289 BELOW val -- no generalization gap for data to close. Pre-registration (b) said the data cell would gain less. It gained more, at p=1.6e-07.

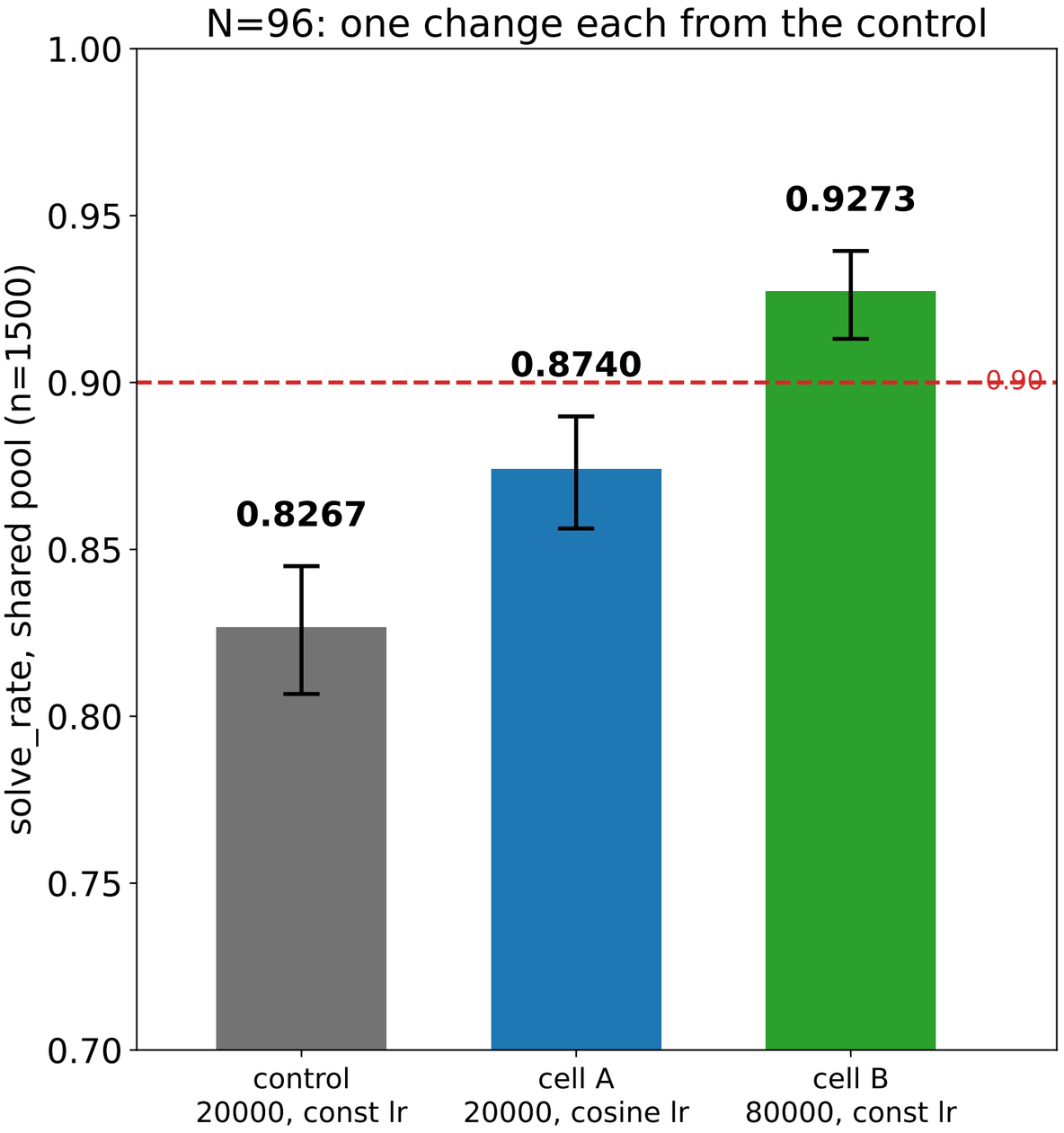
N=48: thin-deep wins at every matched budget, as predicted.

ff=256 beats ff=1024 at all three matched-parameter pairs, +0.5327 / +0.1007 / +0.0480, worst p=3.4e-09. L=8/ff=256 scores 0.9573 -- the best N=48 cell ever, at 1.95x fewer parameters than the previous best.

But the knee is NOT a property of the problem alone.

Narrowing the MLP 4x moves the N=48 knee from L=4 to L=6. Depth and MLP width are partial substitutes, with depth the cheaper currency -- so 'L* is set by the problem' is false as stated in findings 11/14.

Jobs 49565433 (14:41:55) and 49587538 (01:45:31), both ExitCode 0:0. spin_gnn/ at 38d503a, seed=0. Metric of record: shared pool, seed_start=5e10, n=1500, Wilson intervals and paired McNemar exact.



cell	rate	95% Wilson	vs ctrl	p
control, 20k, const	0.8267	[0.8067, 0.8450]	--	--
cell A, 20k, cosine	0.8740	[0.8562, 0.8898]	+0.0473	3.5e-05
cell B, 80k, const	0.9273	[0.9131, 0.9394]	+0.1007	3e-20

Cell B vs cell A, paired on the same 1500 instances:

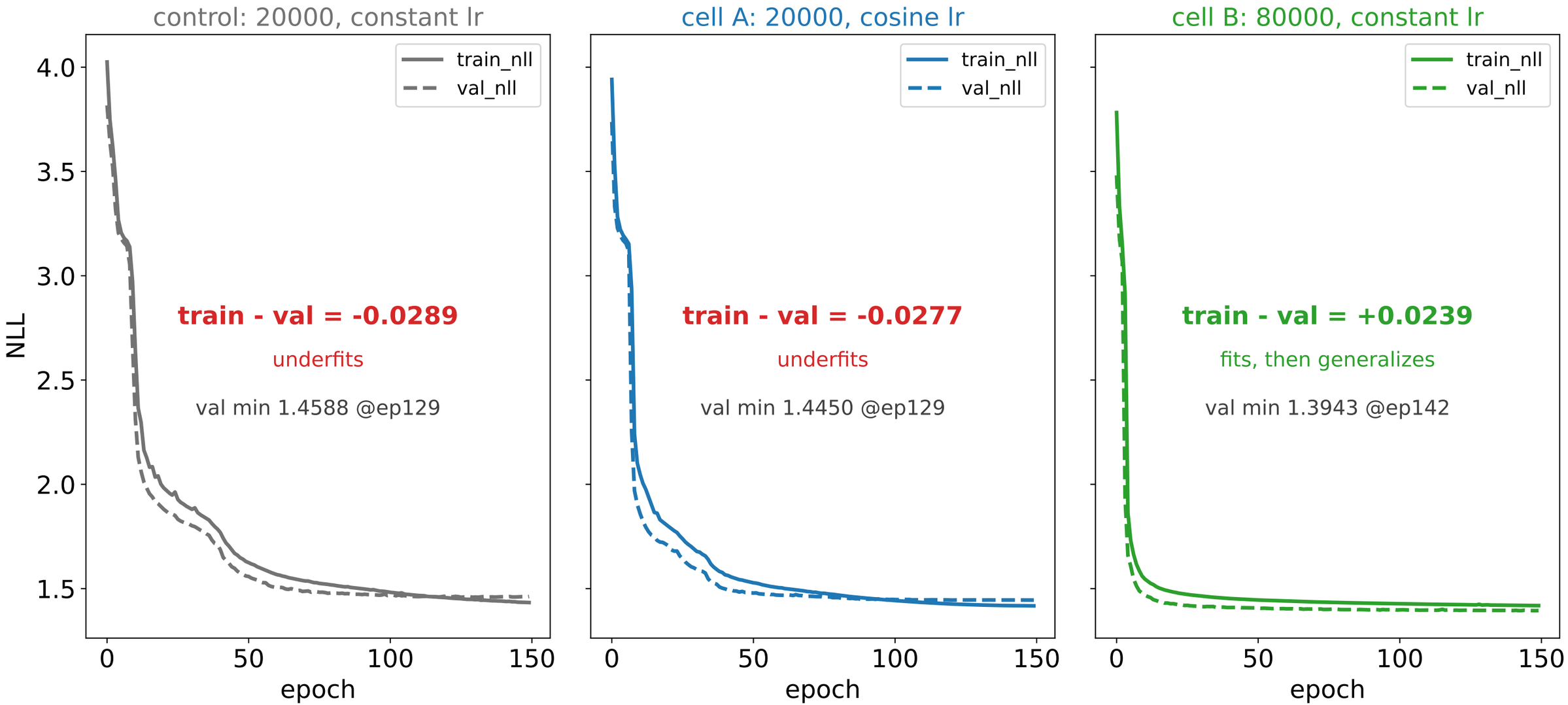
+0.0533, 156-76 discordant, p = 1.6e-07

Pre-registration (b) said B would gain LESS than A.
It gained 2.1x more. The reasoning behind (b) -- that an absent generalization gap implies data-insensitivity -- was wrong, and page 3 shows why.

Pre-registration (c) said neither cell would clear 0.90.
Cell B clears it with the Wilson floor at 0.9131.

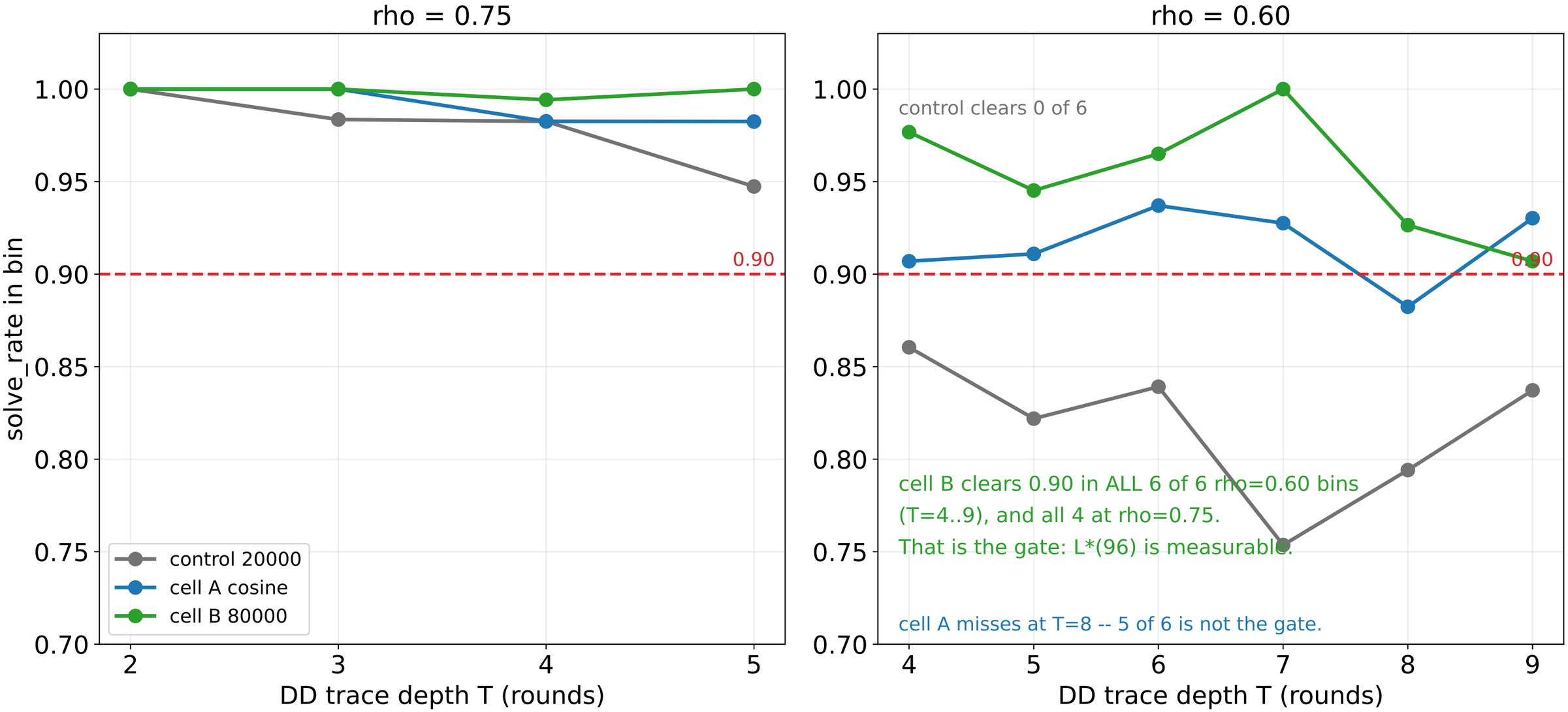
Same 1500 instances for all three cells, so every comparison is paired; deltas are McNemar exact on the discordant pairs, not a test of independent proportions.

Why more data worked: only the 80000 cell stops underfitting

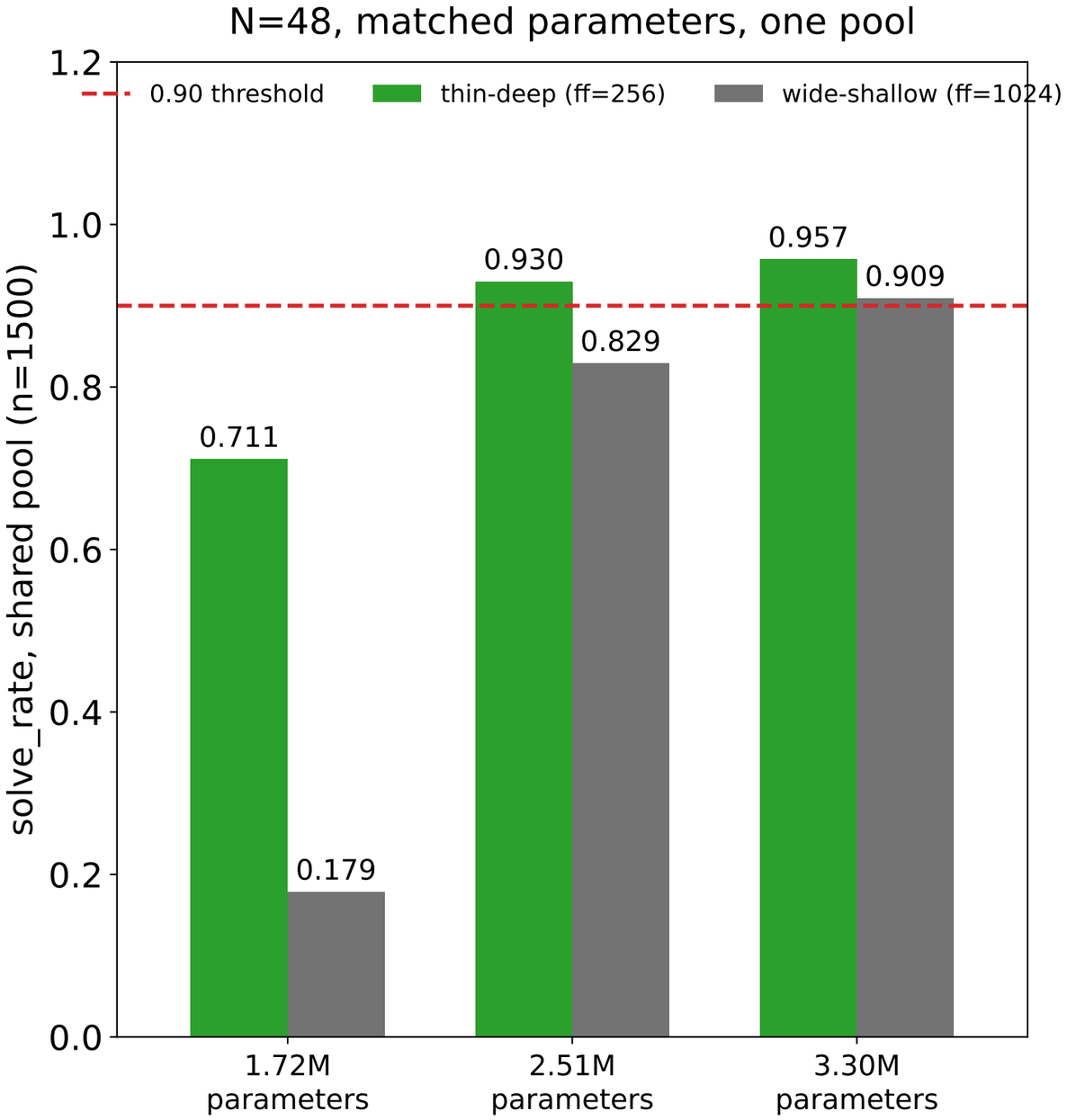


The sign of train - val flips only in cell B. At 20000 the N=96 model never fits its own training set, with or without cosine, so the lever that helps is the one that makes each epoch see more distinct instances. The inference 'no overfitting, therefore more data will not help' does not hold: a model can be data-limited while underfitting.

The gate that unblocks the N=96 layer scan



The N=96 layer scan was gated on a cell clearing 0.90 in EVERY rho=0.60 bin, not merely on aggregate, because an aggregate-only pass buys a column with no L* in it. Cell B passes, so the scan went in as 49639312 at n_train=80000 -- cell B's recipe, one change (num_layers).



budget	thin (256)	wide (1024)	delta	p
1.72M	L=4 0.7113	L=2 0.1787	+0.5327	6.7e-222
2.51M	L=6 0.9300	L=3 0.8293	+0.1007	1e-20
3.30M	L=8 0.9573	L=4 0.9093	+0.0480	3.4e-09

Thin-deep wins all three, replicating finding 13 at N=48.

Discordant pairs (thin solves / wide solves): 811-12, 212-61, 111-39.

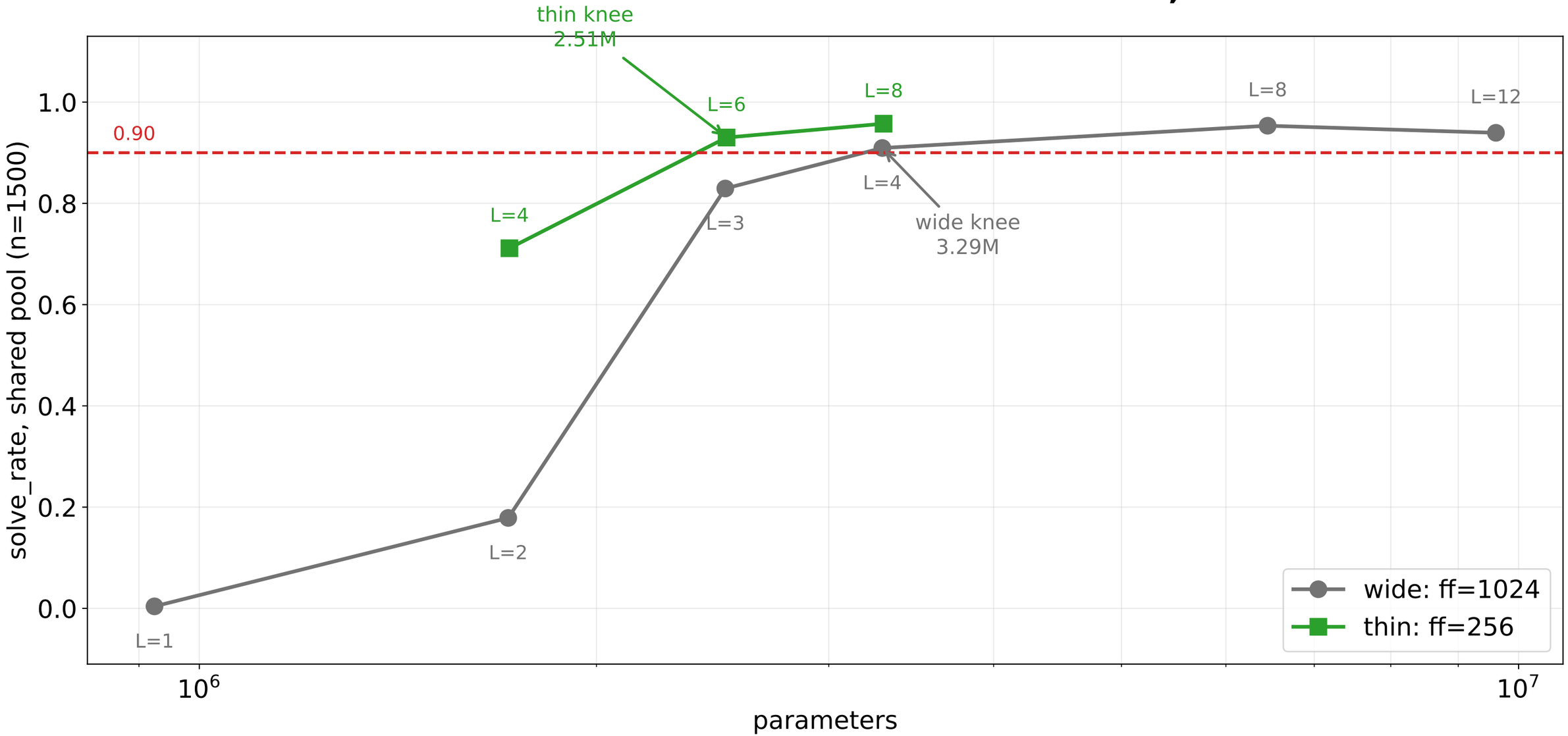
Pre-registration (a), the load-bearing one, tested the pair AT the N=48 knee: L=8/ff=256 vs L=4/ff=1024. +0.0480 at p=3.4e-09, so the thin-deep advantage survives at a size where more depth is actually required. Finding 13 is not an N=24 artifact.

Pre-registration (b) needed > 0.30 at the smallest budget and got +0.5327, matching the +0.4860 seen at N=24.

Measured parameter counts put the thin side only +0.21% ahead (1718074/1714490, 2509626/2504250, 3301178/3294010), so the margins are 23x to 254x the parameter slack. The sbatch header predicted +0.4% from arithmetic; the real slack is tighter.

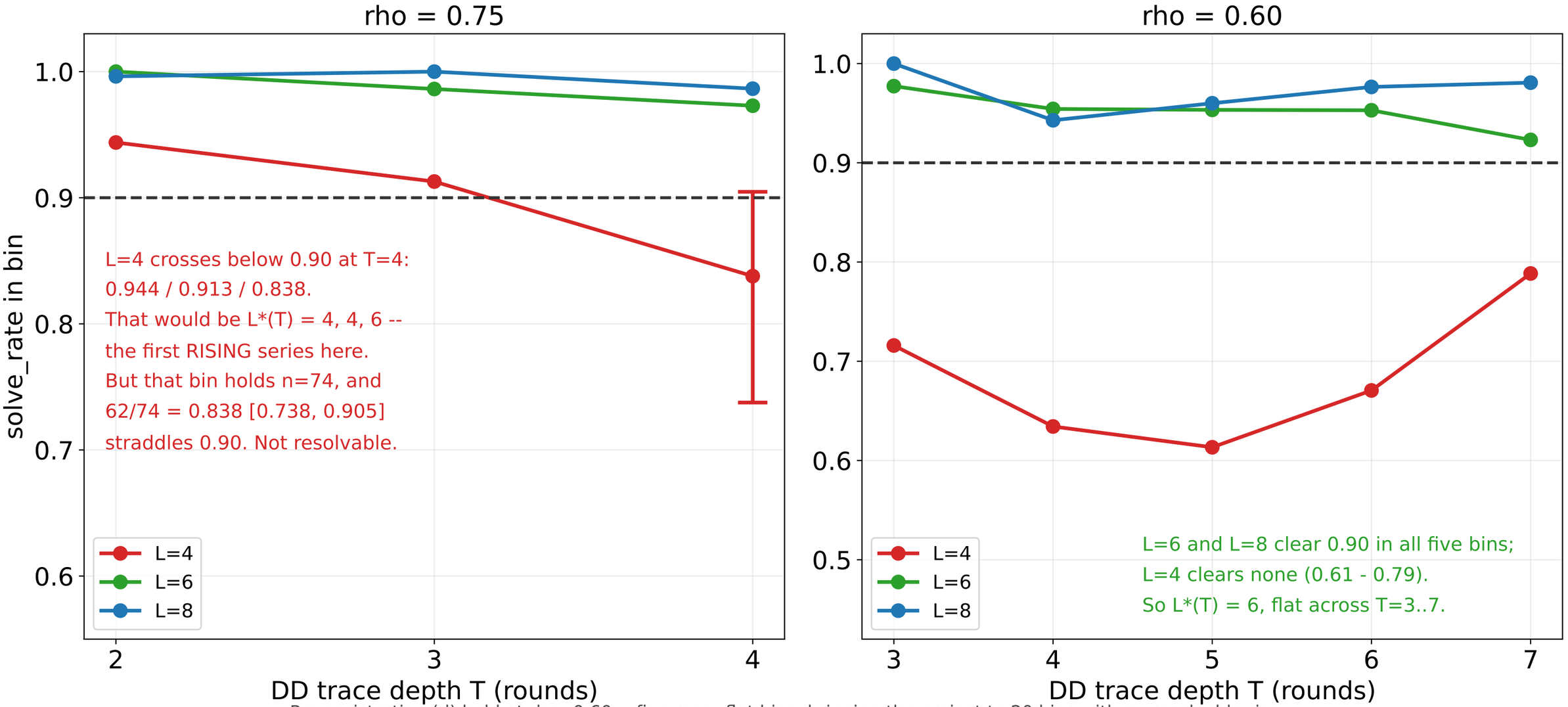
Matched pairs come from 2 thin layers costing 791552 parameters against 1 wide layer at 789760 -- a ratio of 1.9955, both columns sharing a 134970-parameter base, all read off the scan's own n_params column.

The knee moves with MLP width: L=4 wide, L=6 thin



Pre-registration (c) said the thin column's knee would also sit at L=4. It does not: L=4/ff=256 scores 0.7113 and clears 0.90 in none of the five rho=0.60 bins, while L=6/ff=256 clears all five. So L* in LAYERS is not fixed by the problem -- it is 4 at ff=1024 and 6 at ff=256. In PARAMETERS the thin knee is still the cheaper one, 2.51M against 3.29M (+0.0207, p=0.011), which is why depth remains the better currency even though it is not the only one.

L*(T) in the thin column: flat where it is resolvable



Pre-registration (d) held at rho=0.60 -- five more flat bins, bringing the project to 20 bins with no resolvable rise. At rho=0.75 the thin L=4 cell declines monotonically in T and its T=4 point sits below the threshold, which is the shape a rising L* would make; with n=74 the confidence interval still contains 0.90, so it is reported as suggestive and not counted as a rise.

Scorecard: 3 confirmed, 3 falsified, 2 partial

job	pre-registered	verdict	numbers
49565433 (a)	cosine: val min >= 0.020 better AND beats pool	**SPLIT**	val 1.4450 is 0.0138 better, short of 0.020; pool +0.0473 p=3.5e-05
49565433 (b)	4x data gains LESS than cosine	**FALSIFIED**	+0.1007 vs +0.0473; B beats A by +0.0533, p=1.6e-07
49565433 (c)	neither cell clears 0.90	**FALSIFIED**	cell B 0.9273, Wilson floor 0.9131 above 0.90
49565433 (d)	a clearing cell clears EVERY rho=0.60 bin	**CONFIRMED**	cell B clears all 6 (T=4..9) and all 4 at rho=0.75
49587538 (a)	L=8/ff=256 beats the knee, paired p<0.05	**CONFIRMED**	+0.0480, 111-39, p=3.4e-09 -- the load-bearing pair
49587538 (b)	L=4/ff=256 beats L=2/ff=1024 by > 0.30	**CONFIRMED**	+0.5327, 811-12, p=6.8e-222
49587538 (c)	the thin column's knee is also at L=4	**FALSIFIED**	thin knee is L=6; L=4/ff=256 clears no rho=0.60 bin
49587538 (d)	L*(T) stays flat in T at ff=256	**HELD / UNRESOLVED**	flat at rho=0.60 (5 bins); rho=0.75 T=4 straddles 0.90 at n=74

vs the previous best on the SAME eval

N=48 previous best: L=8/ff=1024, 0.9533, 6.45M params.
New best: L=8/ff=256, 0.9573, 3.30M params.
Paired: +0.0040, 57-51, p=0.63 -- a tie on accuracy, at
1.95x fewer parameters. Against L=12/ff=1024 at 9.61M
it is +0.0180, 71-44, p=0.015.
N=96 previous best: 0.8267. New: 0.9273, +0.1007, p=3e-20.
Project best is still 0.9960 at N=24 (job 48912548); the pool
is regenerated per N, so that row shares the protocol only.

provenance

spin_gnn/ package at commit 38d503a, both runs. seed=0;
eval pool seed_start=50000000000, n=1500.
49565433: N=96, d_model=256, nhead=4, L=12, ff=1024, pre-norm,
  lr=1e-3, batch=32, 150 epochs, degree=5, n_colors=4, rho=0.3,
  rho_max=0.95, removal hints. A = cosine; B = 80000 instances.
49587538: N=48, same base, ff=256 at L=4/6/8, constant lr, 20000.
Checkpoints are the FINAL epoch, not the best val (STATUS.md).
Running now: 49639312 (N=96 layer scan at 80000), 49639313
  (N=48 data control at 80000, which removes the budget confound).

Every falsified line was pre-registered with its falsifier written before the run, including (b), where the diagnosis I argued for lost.